

Assignment 1

Analysis of Danish Motor Driven Vehicles over Months

Group 49

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02417 Time Series Analysis

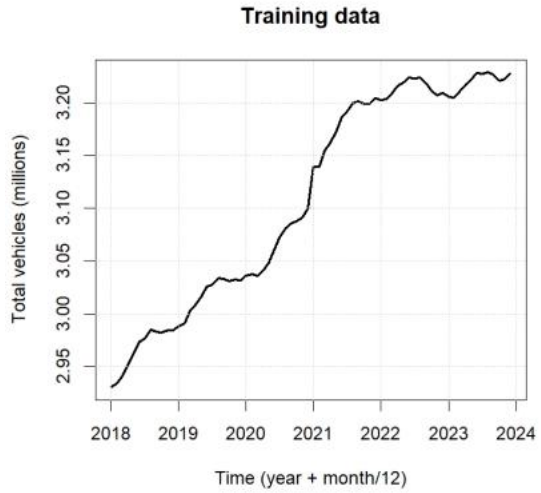
28/02/2026

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1. Plot data

1.1



1.2

The time series shows a clear upward trend from 2018 to 2023. The total number of registered vehicles in Denmark increases steadily over time, rising from approximately 2.93 million in 2018 to about 3.23 million in 2023. The growth is particularly strong around 2020–2021, where the slope becomes steeper. After 2022, the increase slows down and the series appears to level off slightly. There are small short-term fluctuations visible throughout the series, but no very strong seasonal pattern is evident. Overall, the series is dominated by a long-term increasing trend.

2. Linear trend model

2.1

Jiayue Che (s252639)

$$Y_t = x_t\theta + \varepsilon_t$$

$$y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix},\ \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix},\ X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix},\ \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix}\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$$

$$\because Y_1 = 2.930 \quad x_1 = 2018.000$$
$$Y_2 = 2.934 \quad x_2 = 2018.083$$
$$Y_3 = 2.941 \quad x_3 = 2018.167$$

$$\therefore \begin{bmatrix} 2.930 \\ 2.934 \\ 2.941 \end{bmatrix} = \begin{bmatrix} 1 & 2018.000 \\ 1 & 2018.083 \\ 1 & 2018.167 \end{bmatrix}\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$$

3. OLS - global linear trend model

3.1

We estimated the parameters using Ordinary Least Squares (OLS) by minimizing the sum of squared residuals,

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} S(\boldsymbol{\theta})$$

where

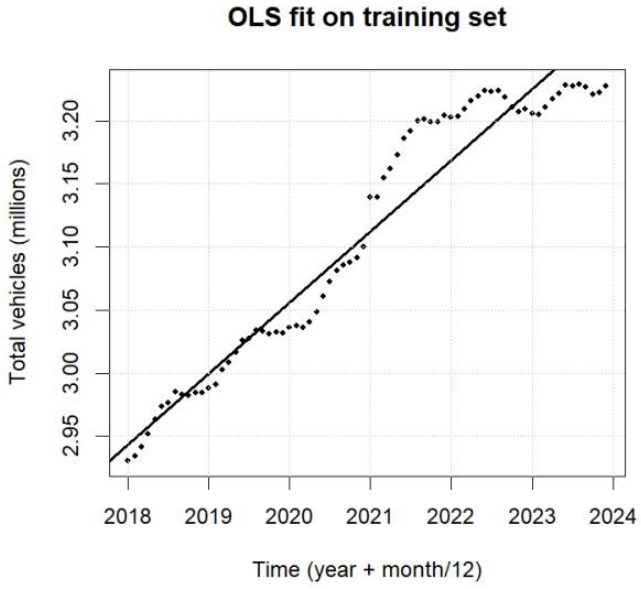
$$S(\boldsymbol{\theta}) = \sum_{t=1}^{n} [y_t - f(\boldsymbol{x}_t; \boldsymbol{\theta})]^2 = \sum_{t=1}^{n} \varepsilon_t^2(\boldsymbol{\theta})$$

which in matrix form gives $\boxed{\widehat{\boldsymbol{\theta}} = (\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{Y}}$ (if $\boldsymbol{x}$ has full rank).

3.2

$\hat{\theta}_1$=-110.35542813； $\hat{\theta}_2$=0.05614456； $\hat{\sigma}_{\hat{\theta}_1}$=3.593581122； $\hat{\sigma}_{\hat{\theta}_2}$=0.001778156

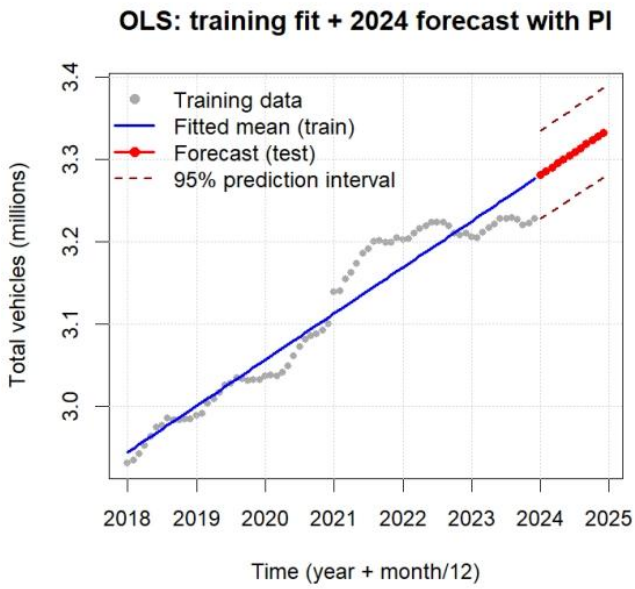
```
> summary(ols_fit)$coefficients # includes standard errors, t-values, p-values
                Estimate  Std. Error    t value     Pr(>|t|)
(Intercept) -110.35542813 3.593581122 -30.70904 2.367907e-42
year           0.05614456 0.001778156  31.57459 3.850121e-43
```



3.3

```
> print(forecast_table)
         time        x    y_obs   y_pred   PI_low  PI_high
73 2024-01-01 2024.000 3.223801 3.281154 3.227579 3.334728
74 2024-02-01 2024.083 3.223805 3.285832 3.232198 3.339467
75 2024-03-01 2024.167 3.231177 3.290511 3.236815 3.344208
76 2024-04-01 2024.250 3.237629 3.295190 3.241430 3.348950
77 2024-05-01 2024.333 3.244611 3.299869 3.246044 3.353693
78 2024-06-01 2024.417 3.255073 3.304547 3.250656 3.358439
79 2024-07-01 2024.500 3.254569 3.309226 3.255267 3.363185
80 2024-08-01 2024.583 3.258060 3.313905 3.259876 3.367934
81 2024-09-01 2024.667 3.256515 3.318583 3.264483 3.372683
82 2024-10-01 2024.750 3.252275 3.323262 3.269090 3.377435
83 2024-11-01 2024.833 3.253059 3.327941 3.273694 3.382188
84 2024-12-01 2024.917 3.258025 3.332620 3.278297 3.386942
```

3.4

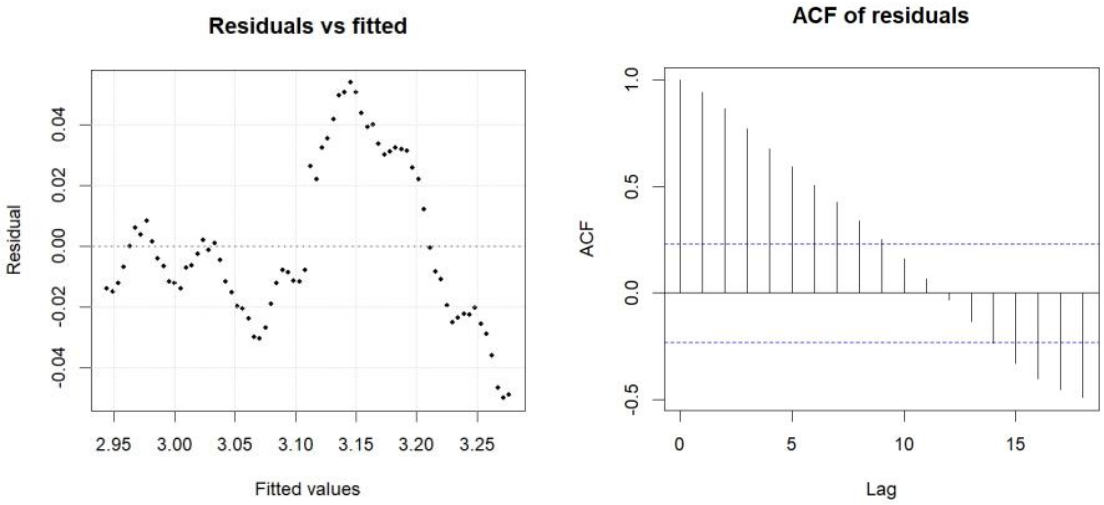


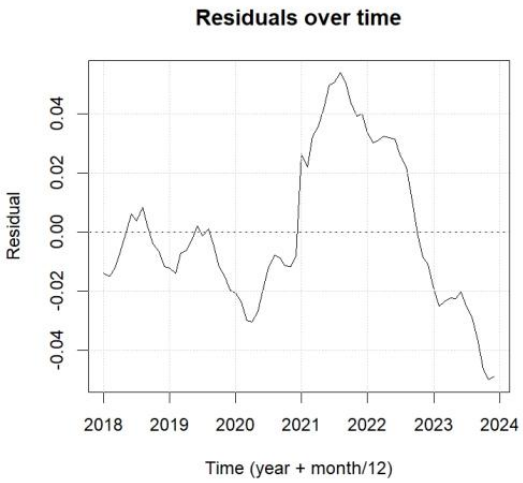
3.5

The RMSE (0.062 million vehicles) corresponds to approximately 2% of the overall mean level (3.13 million vehicles), indicating reasonably accurate point forecasts. However, the 95% prediction interval coverage is only 8.3%, far below the expected 95%. This suggests that the model severely underestimates uncertainty. Overall, although the point forecasts appear acceptable, the model cannot be considered reliable due to poor interval performance.

3.6

The residual diagnostics indicate that the model assumptions are not fulfilled. The residuals over time show clear systematic patterns rather than random fluctuations, suggesting that the linear trend is insufficient. The residuals versus fitted values also display curvature, indicating possible model misspecification. Most importantly, the ACF reveals strong and significant autocorrelation, violating the independence assumption. Overall, the residuals do not resemble white noise, and the OLS trend model is not adequate for this time series.





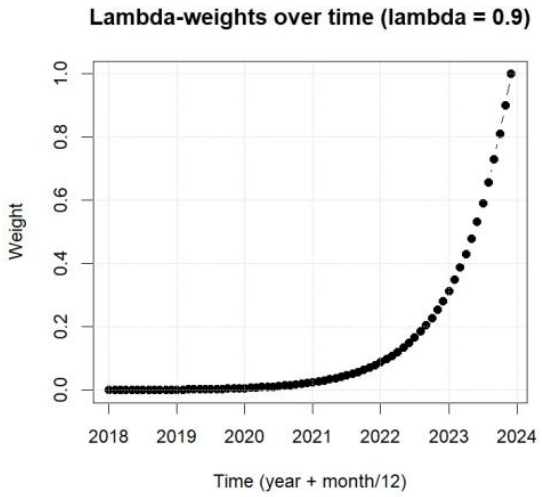
4. WLS - local linear trend model

4.1

In the global OLS model, the error variance–covariance matrix is $\Sigma_{OLS} = \sigma^2 I$, meaning that all observations have equal variance and are uncorrelated. In the local WLS model, the weights decrease geometrically over time according to $w_t = \lambda^{N-t}$. This implies heteroskedastic errors, where $Var(e_t) = \frac{\sigma^2}{w_t}$. Thus, $\Sigma_{WLS} = \sigma^2 \text{diag}(1/w_t)$, which is still diagonal but with unequal variances. Recent observations have smaller error variance (higher weight), while older observations have larger variance (lower weight).

4.2

The λ-weights increase over time and follow an exponential pattern. Since the weights are defined as $w_t=\lambda^{N-t}$, the most recent observation has weight $\lambda^0=1$, which is the highest possible weight. Therefore, the latest time point (2023.917) has the highest weight.



4.3

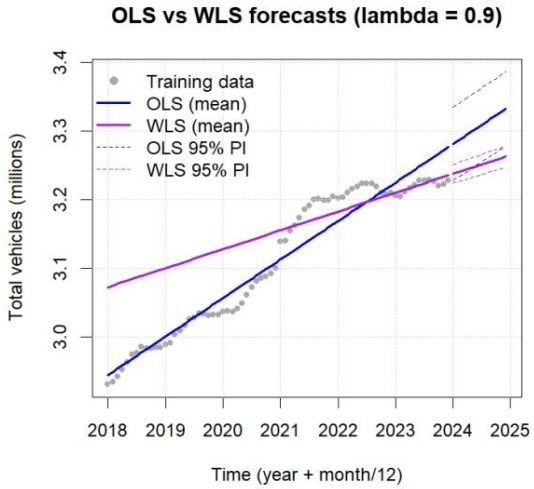
For λ=0.9, the sum of the λ-weights is $\sum w_t=9.9949$. In an OLS model, all observations have equal weight 1, so the corresponding sum of weights is 72.

4.4

```
> coef(wls_fit)  # theta1_hat, theta2_hat (WLS)
(Intercept)        year
-52.4828617   0.0275299
```

4.5

Using the WLS model (λ = 0.9), the forecasts for 2024 increase moderately from approximately 3.251 to 3.279 million vehicles. The prediction standard errors (se_pred) for the WLS forecasts range approximately from 0.0069 to 0.0079. These values are used to construct the 95% prediction intervals. The limited short-term uncertainty around the WLS forecasts. From the plot, the OLS model extrapolates a steeper global trend and produces higher forecasts, whereas the WLS model follows the more recent flattening behaviour of the series. Since recent data suggest slower growth, the WLS forecasts appear more realistic. Therefore, I would choose the WLS predictions.




```
> print(compare_table)
          time     year    y_obs  ols_fit  ols_lwr  ols_upr ols_se_mean ols_se_pred  wls_fit  wls_lwr
73 2024-01-01 2024.000 3.223801 3.281154 3.227579 3.334728 0.006223841  0.02686190 3.237659 3.223937
74 2024-02-01 2024.083 3.223805 3.285832 3.232198 3.339467 0.006353033  0.02689213 3.239953 3.226096
75 2024-03-01 2024.167 3.231177 3.290511 3.236815 3.344208 0.006483038  0.02692314 3.242247 3.228244
76 2024-04-01 2024.250 3.237629 3.295190 3.241430 3.348950 0.006613807  0.02695493 3.244541 3.230381
77 2024-05-01 2024.333 3.244611 3.299869 3.246044 3.353693 0.006745296  0.02698749 3.246836 3.232507
78 2024-06-01 2024.417 3.255073 3.304547 3.250656 3.358439 0.006877464  0.02702083 3.249130 3.234623
79 2024-07-01 2024.500 3.254569 3.309226 3.255267 3.363185 0.007010272  0.02705494 3.251424 3.236729
80 2024-08-01 2024.583 3.258060 3.313905 3.259876 3.367934 0.007143685  0.02708981 3.253718 3.238825
81 2024-09-01 2024.667 3.256515 3.318583 3.264483 3.372683 0.007277670  0.02712545 3.256012 3.240913
82 2024-10-01 2024.750 3.252275 3.323262 3.269090 3.377435 0.007412194  0.02716185 3.258306 3.242991
83 2024-11-01 2024.833 3.253059 3.327941 3.273694 3.382188 0.007547231  0.02719901 3.260601 3.245061
84 2024-12-01 2024.917 3.258025 3.332620 3.278297 3.386942 0.007682751  0.02723693 3.262895 3.247123
    wls_upr wls_se_mean wls_se_pred
73 3.251381 0.002887245 0.006880323
74 3.253810 0.003044814 0.006947917
75 3.256251 0.003208571 0.007021225
76 3.258702 0.003377617 0.007100068
77 3.261164 0.003551196 0.007184265
78 3.263637 0.003728676 0.007273630
79 3.266119 0.003909525 0.007367975
80 3.268611 0.004093296 0.007467111
81 3.271112 0.004279614 0.007570850
82 3.273622 0.004468159 0.007679005
83 3.276140 0.004658662 0.007791393
84 3.278666 0.004850891 0.007907832
```

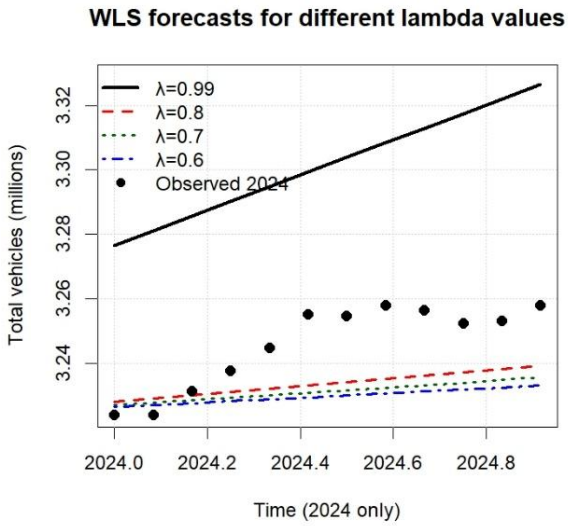
4.6

When λ decreases, the model gives more weight to recent observations and less weight to older data. This leads to increasingly flatter estimated slopes.

From the zoomed plot, it is clear that: For λ = 0.99, the slope is the steepest and the forecasts increase most rapidly. For λ = 0.8, 0.7 and 0.6, the slopes gradually decrease. For λ = 0.6, the forecast is the flattest and relies almost entirely on recent data.

Since the most recent observations in 2023–2024 show slower growth compared to the long-term trend, smaller λ values produce more conservative and realistic forecasts.

Thus, the slopes correspond well to theoretical expectations: the smaller the λ, the more local the model, and the flatter the forecast.



5. Recursive estimation and optimization of λ

5.1

update equations:

For $\hat{\theta}_t$: $\hat{\theta}_t = \hat{\theta}_{t-1} + \frac{R_{t-1}\cdot\varphi_t}{1+\varphi_t^T\cdot R_{t-1}\cdot\varphi_t}\cdot(y_t - \varphi_t^T\cdot\hat{\theta}_{t-1})$

For R_t: $R_t = R_{t-1} - \frac{R_{t-1}\cdot\varphi_t}{1+\varphi_t^T\cdot R_{t-1}\cdot\varphi_t}\cdot R_{t-1}\cdot\varphi_t^T$

$\because \varphi_t^T\cdot R_0\cdot\varphi_1 = [1\ \ 2018]\begin{bmatrix}0.1 & 0\\ 0 & 0.1\end{bmatrix}\begin{bmatrix}1\\ 2018\end{bmatrix} = 0.1\times(1^2+2018^2) = 407232.5$

$R_0\cdot\varphi_1\cdot\varphi_1^T\cdot R_0 = \begin{bmatrix}0.1 & 0\\ 0 & 0.1\end{bmatrix}\begin{bmatrix}1\\ 2018\end{bmatrix}[1\ \ 2018]\begin{bmatrix}0.1 & 0\\ 0 & 0.1\end{bmatrix}$

$= \begin{bmatrix}0.1\\ 201.8\end{bmatrix}[1\ \ 2018]\begin{bmatrix}0.1 & 0\\ 0 & 0.1\end{bmatrix}$

$= \begin{bmatrix}0.1 & 201.8\\ 201.8 & 407232.4\end{bmatrix}\begin{bmatrix}0.1 & 0\\ 0 & 0.1\end{bmatrix}$

$= \begin{bmatrix}0.01 & 20.18\\ 20.18 & 40723.24\end{bmatrix}$

$\therefore R_1 = R_0 - \frac{R_0\varphi_1\varphi_1^T R_0}{407233.5} \approx \begin{bmatrix}9.999\,997\,54\times10^{-2} & -4.955\,388\,00\times10^{-5}\\ -4.955\,388\,00\times10^{-5} & 2.701\,153\,02\times10^{-7}\end{bmatrix}$

$R_2 = R_1 - \frac{R_1\varphi_2\varphi_2^T R_1}{1+\varphi_2^T R_1\varphi_2} \approx \begin{bmatrix}9.999\,997\,54\times10^{-2} & -4.955\,291\,77\times10^{-5}\\ -4.955\,291\,77\times10^{-5} & 1.473\,297\,16\times10^{-7}\end{bmatrix}$

5.2

We implemented the RLS update equations in a for-loop for the linear trend model $y_t = \theta_1 + \theta_2 x_t + \varepsilon_t$, using $\varphi_t = [1, x_t]^T$. The initialization was $R_0 = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}$ and $\hat{\theta}_0 = [0, 0]^T$. At each time step we computed the gain

$$K_t = \frac{R_{t-1}\varphi_t}{1 + \varphi_t^T R_{t-1}\varphi_t},$$

updated the parameters

$$\hat{\theta}_t = \hat{\theta}_{t-1} + K_t(y_t - \varphi_t^T \hat{\theta}_{t-1}),$$

and updated R_t accordingly. Using the first three observations we obtained:

$$\hat{\theta}_1 = \begin{bmatrix} 7.196075 \times 10^{-7} \\ 0.001452168 \end{bmatrix}, \hat{\theta}_2 = \begin{bmatrix} 7.129135 \times 10^{-7} \\ 0.001453022 \end{bmatrix}, \hat{\theta}_3 = \begin{bmatrix} 6.765664 \times 10^{-7} \\ 0.001454506 \end{bmatrix}.$$

Although the matrix expressions can look complex, the recursion is intuitive: at each step we form a prediction, compute the prediction error, and adjust the parameters by a gain that reflects how informative the new data point is.

5.3

We computed the Ordinary Least Squares (OLS) estimates on the full training set for the linear trend model

$$y_t = \theta_1 + \theta_2 x_t + \varepsilon_t.$$

The OLS estimates were:

$$\hat{\theta}^{OLS} = \begin{bmatrix} -110.3554 \\ 0.0561446 \end{bmatrix}.$$

Using Recursive Least Squares (RLS) with $\lambda = 1$, and initialization

$$R_0 = 0.1I, \hat{\theta}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

the final RLS estimate at time $t = N$ was:

$$\hat{\theta}_N^{RLS} = \begin{bmatrix} -0.0005827 \\ 0.0015394 \end{bmatrix}.$$

The difference between RLS and OLS was therefore:

$$\hat{\theta}_N^{RLS} - \hat{\theta}^{OLS} = \begin{bmatrix} 110.3548 \\ -0.0546052 \end{bmatrix}.$$

The two estimates are clearly not close.

In theory, with λ = 1, RLS should converge to OLS.

The discrepancy is caused by the small initial covariance R_0, which leads to small gains and slow adaptation.

Increasing R_0gives more weight to incoming data and reduces the difference.

Thus, proper initialization is crucial for convergence.

5.4

We implemented Recursive Least Squares with forgetting using two values of the forgetting factor: $\lambda = 0.7$and $\lambda = 0.99$.

The update equations were:

$$K_t = \frac{R_{t-1}\varphi_t}{\lambda + \varphi_t^T R_{t-1}\varphi_t}$$

$$\hat{\theta}_t = \hat{\theta}_{t-1} + K_t(y_t - \varphi_t^T \hat{\theta}_{t-1})$$

$$R_t = \frac{1}{\lambda}\left(R_{t-1} - \frac{R_{t-1}\varphi_t\varphi_t^T R_{t-1}}{\lambda + \varphi_t^T R_{t-1}\varphi_t}\right)$$

The parameter trajectories were plotted after removing a short burn-in period.

For $\lambda = 0.7$, the final parameter estimates were:

$$\hat{\theta}_N^{(0.7)} = \begin{bmatrix} -15.71469 \\ 0.00935849 \end{bmatrix}$$

For $\lambda = 0.99$, the final estimates were:

$$\hat{\theta}_N^{(0.99)} = \begin{bmatrix} -0.00081270 \\ 0.00154905 \end{bmatrix}$$

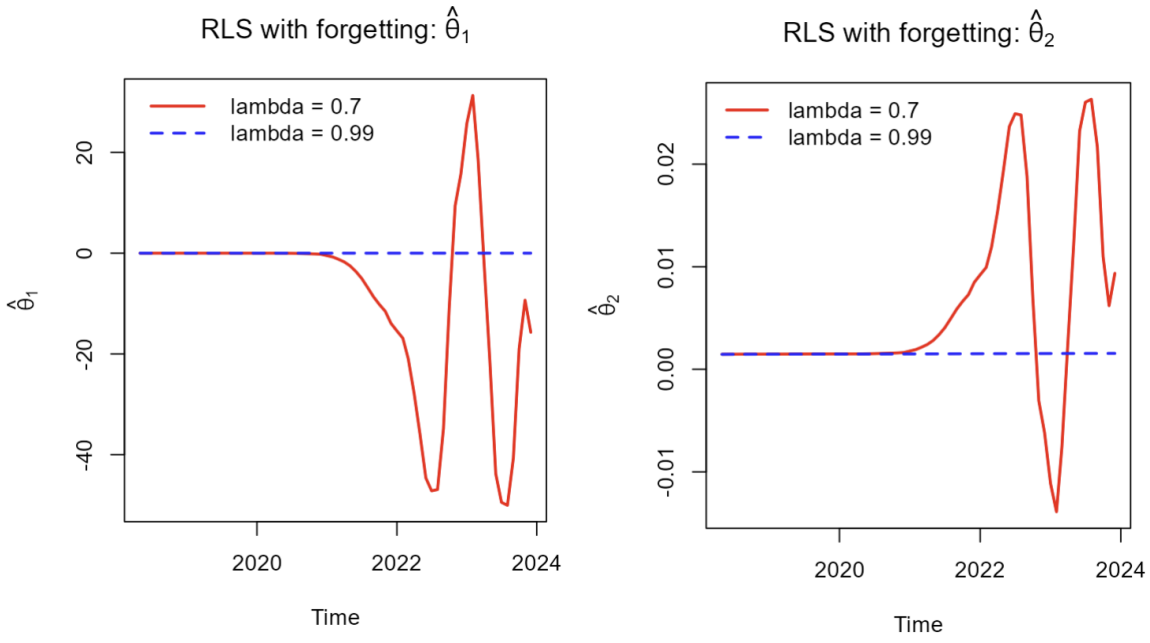
Observations from the parameter plots:

- λ = 0.99 → smooth and stable estimates, close to global trend.
- λ = 0.7 → strong fluctuations and high volatility.

Smaller λ gives short memory and high adaptivity.

Larger λ gives long memory and stable behavior.

Thus, the forgetting factor controls the trade-off between stability and adaptivity in time-adaptive models.



5.5

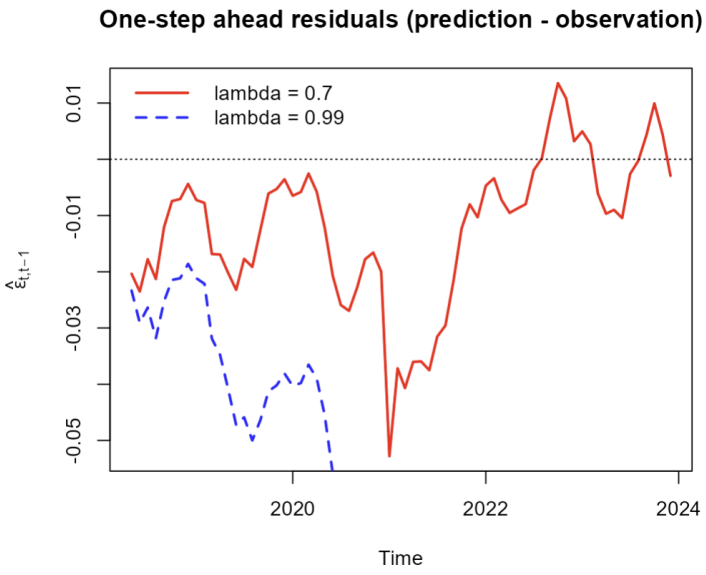
One-step residuals show:

- $\lambda = 0.7 \rightarrow$ larger fluctuations and stronger reactions to local changes.
- $\lambda = 0.99 \rightarrow$ smoother residuals with smaller variance.

Thus:

- Small $\lambda \rightarrow$ adaptive but volatile
- Large $\lambda \rightarrow$ stable but slower adaptation

This illustrates the stability–adaptivity trade-off.



5.6

The RMSE curves show a clear and consistent pattern across all horizons $k = 1, \ldots, 12$.

For every prediction horizon, the RMSE increases as λ increases.

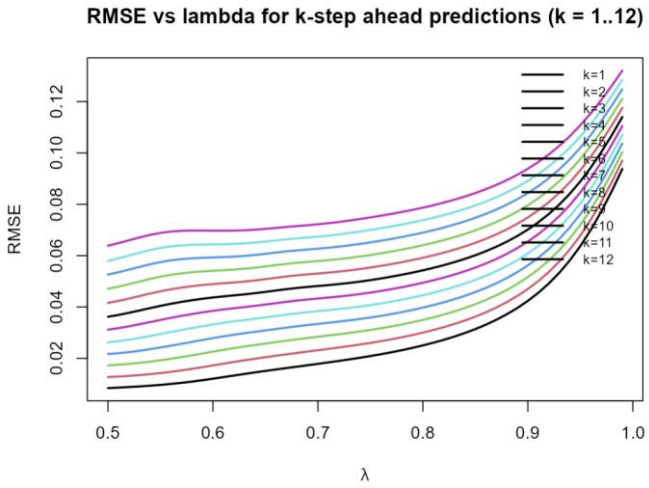
The minimum RMSE is achieved at the smallest values of λ (around 0.5–0.6), and the prediction error grows steadily as λ approaches 1.

This indicates that, for this dataset, a more adaptive model (smaller λ) provides better predictive performance. Since the time series exhibits noticeable changes in trend over time, giving more weight to recent observations improves forecast accuracy.

Furthermore, the relative shape of the RMSE curves is similar for all horizons. Although the overall RMSE level increases with the prediction horizon (as expected), the optimal λ does not vary substantially across different values of k.

Therefore, there is no strong evidence that λ needs to depend on the forecast horizon. A single moderately small value of λ (around 0.5–0.6) appears to provide good performance across all horizons.

Overall, the results highlight the trade-off between stability and adaptivity: in this case, adaptivity dominates, and smaller values of λ yield better forecasts.



5.7

The OLS forecast clearly overestimates the 2024 values. Since OLS fits a global linear trend to the entire 2018–2023 period, it extrapolates the earlier strong upward trend too aggressively. As a result, the predicted values lie well above the observed test data.

The RLS forecast with $\lambda = 0.55$ underestimates the 2024 values. Because the forgetting factor is relatively small, the model gives high weight to the most recent observations. This leads to a flatter estimated slope at the end of the training period and therefore

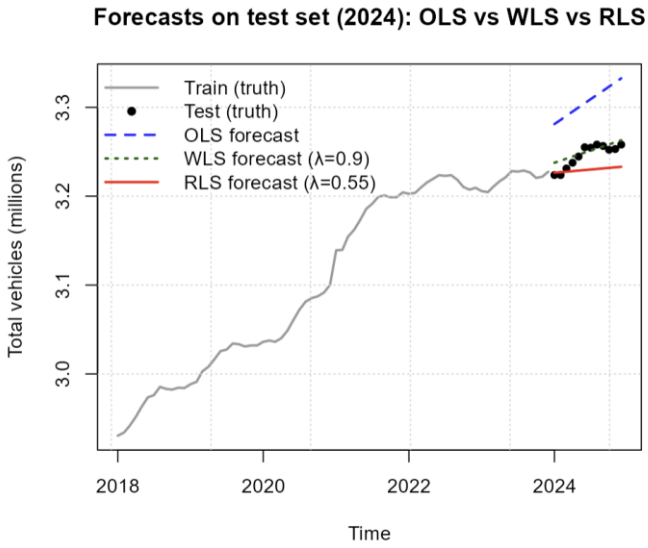
lower forecasts for 2024.

The WLS forecast ($\lambda = 0.9$) appears to track the test data more closely than both OLS and RLS. It provides a compromise between global stability and local adaptivity, resulting in predictions that align better with the observed values in 2024.

Overall, the comparison illustrates the trade-off between global and adaptive models:

- OLS: stable but too rigid, overestimates future values
- RLS (small λ): highly adaptive but may underreact or overreact depending on recent fluctuations
- WLS: intermediate behavior, providing a balanced forecast in this case

Thus, for the 2024 test set, the weighted model with moderate forgetting performs best visually.



5.8

(1) Overfitting vs. underfitting

The forgetting factor λ controls a bias–variance trade-off.

Small λ → highly adaptive but risk of overfitting.

Large λ → stable but risk of underfitting structural changes.

(2) Challenges with test sets

Time series data are serially dependent, so random splitting causes information leakage. Test data must be future observations, which reduces training size and makes results split-dependent.

(3) Role of recursive estimation

Recursive estimation mimics real-time updating and fits naturally with rolling

evaluation.

However, it does not replace proper out-of-sample validation.

(4) Other adaptive techniques

Examples include rolling window estimation, exponentially weighted LS, state-space models, Kalman filtering, and Bayesian dynamic models.

(5) Additional thoughts

Time-adaptive models handle structural changes well but are sensitive to λ and may increase forecast volatility. Careful tuning is essential.